

RESEARCH

Open Access



N-acetylcysteine reduces serum creatinine, blood urea nitrogen, symmetric dimethylarginine and urine protein to creatinine ratio in cats with chronic kidney disease: a double-blind, placebo-controlled clinical trial

Hadi Alihosseini¹, Ekrem Çagatay Çolakoğlu^{2*}, Ali Evren Haydardedeoğlu³ and Doğukan Özen⁴

Abstract

Background Oxidative stress is considered a significant contributing factor of chronic kidney disease (CKD). To date, there is a paucity of clinical data in the literature regarding the effect of N-Acetylcysteine (NAC) in cats with naturally developing CKD. The aim of the study is to evaluate whether the addition of NAC in the treatment of cats with acute exacerbations of CKD could improve kidney function biomarkers over the use of intravenous fluid therapy alone.

Methods A total of 50 client-owned cats were included in the study. The inclusion criteria comprised cats previously diagnosed with azotemic CKD (IRIS stage 2–4) in addition to ultrasonographic evidence of bilaterally decreased renal mass, rough surface contours, and alteration of renal cortical echogenicity. All cats were examined using standard clinical procedures, including clinical examination, blood analyses, abdominal ultrasonography, dipstick urinalysis and urine culture. Computer-generated randomisation was utilised to assign the cats into the following groups: NAC ($n:40$): N-acetylcysteine (70 mg/kg, diluted in 50 ml 0.9% saline solution, administered intravenously over a period of seven days, and a placebo group ($n:10$) 50 ml 0.9% saline solution, IV for 7 days. Blood analyses and dipstick urinalysis were repeated on the eighth day of treatment. Between-group differences in baseline age and weight were assessed using the Student's t-test, while sex distribution was evaluated with the Fisher's exact test. Treatment effects across time were analysed using a two-way mixed-design ANOVA, with "Group" and "Time" entered as fixed factors and their interaction term included in the model.

Results SDMA and creatinine concentrations decreased significantly in both groups, but the concentrations of both were significantly lower in the NAC group after treatment (Day 8 values: SDMA NAC 16.5 ± 1.21 µg/dl versus placebo

*Correspondence:
Ekrem Çagatay Çolakoğlu
colakoglu@ankara.edu.tr

Full list of author information is available at the end of the article



© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

$27 \pm 3.89 \mu\text{g/dl}$; $P=0.04$ and Creatinine NAC $4.01 \pm 0.25 \text{ mg/dl}$ versus placebo $6.44 \pm 0.9 \text{ mg/dl}$; $P < 0.001$). UPC and BUN decreased significantly in the NAC group, but no change was observed in the placebo group.

Conclusion The incorporation of NAC into treatment regimens demonstrates potential as a treatment strategy for cats with acute-on-chronic kidney disease.

Keywords Acetylcysteine, Cat, Kidney disease, Oxidative stress

Background

Chronic kidney disease (CKD) has been reported as a prevalent clinical condition in geriatric cats [1]. While the precise pathophysiology of CKD remains to be fully elucidated, the presence of cellular injury has been established, thereby initiating a process in which oxidative stress emerges as a potential pivotal factor in the progression of CKD [2]. Oxidative stress is characterised by an imbalance between prooxidative processes and antioxidant defense mechanisms. It is also the result of the production of reactive oxygen species (ROS). The deterioration of antioxidant defense mechanisms is known to result in damage to biological macromolecules, as well as the disruption of metabolism and physiology of the tissues [3]. Kidney tissue cells are characterised by an abundance of mitochondria, rendering them particularly susceptible to oxidative stress. In particular, tubular cells are characterised by their high metabolic activity and exposure to elevated levels of ROS [4]. In human patients, increased ROS production has been observed to be associated with proteinuria and the progression of CKD [5]. Furthermore, cats with CKD demonstrate the presence of oxidative stress [6]. The primary function of N-Acetylcysteine (NAC) is associated with its antioxidant and anti-inflammatory properties. NAC has been demonstrated to have a plethora of biological functions, including the scavenging of oxygen free radicals, the regulation of cell metabolism by preventing DNA damage and the regulation of gene expression [7, 8]. It is hypothesised that NAC supplementation may reduce nerve damage, most likely due to the antioxidant, anti-inflammatory, and anti-apoptotic effects of NAC [9]. The renoprotective effects of NAC in the prevention of contrast-induced nephropathy and the improvement of kidney function have been reported in human patients [10]. A number of studies have reported a decrease in serum creatinine after exposure to NAC. In experimental and human studies exploring the potential for kidney damage reduction, NAC has been observed to curtail the escalation of serum creatinine, BUN, and proteinuria [11–13]. In the field of animal studies, the administration of NAC has been documented to attenuate the effects of ischemic kidney failure in glomerular mesangial cells [14]. In rat models of kidney failure induced by the administration of NSAIDs NAC has been demonstrated to effectively alleviate the impairment of glomerular function by means

of kidney vasodilatory induction, whilst concurrently suppressing oxidative stress through the obstruction of intrarenal reactive oxygen storage [15]. A review of the literature reveals a paucity of clinical data concerning the use of NAC in cats with CKD. The objective of the present study is to evaluate whether the addition of NAC in the treatment of cats with acute exacerbations of CKD could improve kidney function biomarkers over the use of intravenous fluid therapy alone.

Materials and methods

Animals

Non-pregnant and normotensive client-owned cats were referred with signs of chronic kidney disease (CKD), including vomiting, polyuria, polydipsia, anorexia, halitosis, lethargy, and small kidneys on palpation. The inclusion criteria encompassed the evaluation of IRIS stage 2, 3, and 4 cats [16] in conjunction with confirmed ultrasonographic evidence of bilaterally decreased renal mass, rough surface contours, and alteration of renal cortical echogenicity. Cats were excluded from the study if they had pre-renal (i.e. heart disease, hepatic disorders, neoplastic conditions, acute dehydration, diabetes mellitus and hyperthyroidism) or post-renal pathologies (obstructive uropathies) affecting kidney function. Cats with renal cysts, nephrolith and pyelonephritis were excluded from the study. Cases with cystitis were also excluded due to its possible effects on UPC. Furthermore, cats diagnosed with feline infectious peritonitis, feline immunodeficiency virus or feline leukemia virus by rapid tests were excluded from the study. It is also noteworthy that none of the cats had received medical or surgical treatment at the time of referral or within the preceding three months prior to admission. The final total of 50 cats included in the study was determined based on these inclusion and exclusion criteria. The cats enrolled in the study had an IRIS classification performed in the same hospital in the stable phase of CKD. All cats were being fed commercial renal dry food before admission and had undergone regular antiparasitic therapy and routine vaccinations. Written consent was also obtained from the owners. The present study was carried out in accordance with the principles of the European Convention for the Protection of Vertebrate Animals Used for Experimental and Other Scientific Purposes and the requirements of European

Union Directive 2010/63 on the protection of animals used for experimental and other scientific purposes.

Study design

In this prospective, randomised, placebo-controlled trial, all cats were examined using a series of standard clinical procedures. These included a complete clinical examination, as well as blood analyses (CBC, glucose, creatinine, blood urea nitrogen, total protein, albumin, inorganic phosphorus, alanine aminotransferase, alkaline phosphatase, aspartate aminotransferase, symmetric dimethylarginine-SDMA, total thyroxine, sodium, potassium, calcium, chloride [IDEXX procyte hematology and catalyst one Blood Chemistry Analyzer]), abdominal ultrasonography, dipstick urinalysis, urine culture and urine protein to creatinine ratio (UPC). Thyroid hormone levels were measured in all cats included in the study, and those diagnosed with hyperthyroidism were excluded. The mean of five to seven consecutive consistent indirect blood pressure measurements was obtained with an oscillometric device (Vet-Dop2, Veterinary Doppler blood pressure system, Vmed technology) [17]. Computer-generated randomisation was utilised to assign the cats into the following groups: NAC ($n:40$): N-acetylcysteine (70 mg/kg, diluted in 50 ml 0.9% saline solution, administered intravenously over a period of 7 days [18, 19], and a placebo group ($n:10$) 50 ml 0.9% saline solution, IV for 7 days). Fluid therapy with ringer solution was also provided intravenously to all cats, with the dosage calculated based on the patient's body weight (kg) and their dehydration status (as a decimal), as determined by physical examination parameters [20]. All cats in study had a dehydration status as 6%–8% [21]. All therapeutic interventions were administered twice daily for a period of one week at the hospital. All cats in groups received no other medications during the study, despite the study protocol allowing the administration of ancillary medications. A 7-day course of treatment was deemed adequate for cats that refused voluntary food intake and showed standardized dehydration levels of 6–8%. Water and dry food were offered ad libitum to all patients throughout the hospitalization period but none of the cats consumed them voluntarily. Force oral feeding was not also preferred due to risk of inducing vomiting. Blood analyses were repeated on the eighth day of treatment. The treatment protocol utilised in the groups was presented in a uniform physical manner to both researchers and owners. Treatment was also continued for CKD cats that still required treatment after seven days. Subsequent to the 7-day treatment period, a special prescription dry or wet kidney diet were continued in all CKD cats that commenced oral intake.

Statistical analysis

The descriptive statistics for each variable were calculated and presented as “Mean $\pm$ Standard Error of Mean”. Prior to conducting the statistical analysis, the data underwent scrutiny using the Shapiro-Wilk test for normality and the Levene test for homogeneity of variances, as these are fundamental to the validity of parametric test assumptions. The Student t-test was employed to evaluate the differences between the study groups with respect to age and weight. The Fisher exact test was used to assess the frequency distribution of sex among study groups. The collected data were subjected to two-factor mixed-design ANOVA (analysis of variance) using General Linear Model procedure for repeated measurements. The model incorporated “Group” and “Time” as the primary effects, along with their two-way interaction terms. Post hoc testing was performed using simple effect analysis with Bonferroni adjustment. To align scientific rigor with clinical responsibility, we implemented a computer-generated randomization scheme that allowed for unbalanced allocation. Unbalanced randomization is accepted in both human and veterinary clinical trials, particularly when greater information is needed on the efficacy or safety of the new treatment, or ethical/practical constraints limit placebo allocation [22, 23]. From a statistical perspective, our design is supported by power analysis. Based on using Cohen's $f = 0.25$, α (type 1 error rate) = 0.05 and 80% power in a two way mixed design ANOVA, the required total sample size was approximately 34 subjects [24]. A probability value of less than 0.05 was considered to be significant, unless otherwise indicated. The statistical analysis was conducted using SPSS 21.

Results

A total 50 of 58 client-owned cats were included in the study, comprising Chinchila ($n:1$), domestic short hair ($n:19$), Persian ($n:5$), Ankara cat ($n:4$), Siamese ($n:9$), Scottish fold cat ($n:7$), British short hair ($n:3$), Himalayan ($n:1$) and Van cat ($n:1$). Eight cats were excluded from the study for the following reasons: previous drug therapy ($n:3$), concurrent renal mass ($n:2$), renal cysts ($n:1$) and renal calculi ($n:2$). The mean age, weight and sex distributions of cats in the groups are shown in Table 1. Prior to treatment, serum creatinine, SDMA, BUN and UPC levels were higher in the groups (Table 2). SDMA and creatinine concentrations decreased significantly in both groups, but the concentrations of both were significantly lower in the NAC group after treatment (Day 8 values: SDMA NAC 16.5 ± 1.21 $\mu\text{g/dl}$ versus placebo 27 ± 3.89 $\mu\text{g/dl}$; $P=0.04$ and Creatinine NAC 4.01 ± 0.25 mg/dl versus placebo 6.44 ± 0.9 mg/dl ; $P<0.001$). UPC and BUN decreased significantly in the NAC group, but no change was observed in the placebo group (Table 2). Furthermore, the tables present the mean values of RBC, HCT,

Table 1 Characteristics of cats in groups

	NAC Group (n:40)	Placebo Group (n:10)	P
Age (years)	10.74 ± 0.53	11.5 ± 0.75	0,501
Weight (kg)	3.97 ± 0.14	4.21 ± 0.25	0,428
Sex (n, %)	♀: 24 (60) ♂: 16 (40)	♀: 6 (60) ♂: 4 (40)	
Breeds (n)	Chinchila (n:1) Domestic short hair (n:17) Persian (n:4) Ankara cat (n:2) Siamese (n:7) Scottish fold cat (n:5) Himalayan (n:1) British short hair (n:2) Van cat (n:1)	Domestic short hair (n:2), Persian (n:1), Ankara cat (n:2), Siamese (n:2), Scottish fold cat (n:2), British short hair (n:1)	
IRIS Stage 2	9	3	
IRIS Stage 3	18	5	
IRIS Stage 4	13	2	

NAC N-acetylcysteine, IRIS International Renal Interest Society

HGB, PHOS, TP and ALB. A comparative dipstick urinalysis of urine gravity between the NAC and control groups on day 8 revealed a range from 1.026 to 1.029 and from 1.024 to 1.032, respectively, with no significant relationship observed ($P > 0.05$). Culture results revealed no pathogens in all cats. All cats survived to discharge from

the hospital. Discharge times ranged from 8 to 12 days for the NAC group and 9–14 days for the placebo group. No adverse effects associated with NAC administration were also observed.

Discussion

The antioxidative and anti-inflammatory properties of NAC render it a promising therapeutic agent against the destructive effects of oxidative stress. Indeed, it is one of the most widely used antioxidants in human medicine today [25]. Numerous studies have demonstrated the efficacy of NAC in the treatment of a wide range of diseases, including diabetes, obesity, cancer, cardiovascular diseases, cardiovascular problems and autoimmunity [26]. A substantial number of clinical research articles and studies have emphasised the beneficial therapeutic effects of NAC in the veterinary field, including the following: investigation of the pharmacokinetic effects of NAC in cats and chickens [27, 28], protective effects in dogs exposed to acetaminophen toxicity [29], healing of infectious keratitis wounds in cats [30], proven effects of NAC on strengthening the intestinal mucosal barrier in experimental intestinal diseases of pigs [31], optimal therapeutic effects in mastitis and metritis of cows [32],

Table 2 Blood analyses in groups

	Group	t0		t8		P		
		Mean ± SEM	SD	Mean ± SEM	SD	Group (G)	Time (T)	G × T
RBC	NAC Group	8.16 ± 0.23 ^{a, A}	1,46	6.68 ± 0.24 ^{b, B}	1,54	0,004	< 0.001	0,435
	Placebo Group	9.75 ± 0.48 ^{b, A}	1,53	7.81 ± 0.56 ^{a, B}	1,78			
HCT	NAC Group	34.34 ± 1.13 ^{b, A}	7,12	27.58 ± 1.01 ^{b, B}	6,39	0,002	< 0.001	0,349
	Placebo Group	41.53 ± 2.21 ^{a, A}	6,99	31.78 ± 2.51 ^{a, B}	7,95			
HGB	NAC Group	11.44 ± 0.31 ^{b, A}	1,97	9.94 ± 0.36 ^{b, B}	2,29	0,027	< 0.001	0,442
	Placebo Group	13.11 ± 0.56 ^{a, A}	1,78	10.83 ± 0.74 ^{a, B}	2,35			
SDMA	NAC Group	30.65 ± 2.48 ^{a, A}	15,7	16.5 ± 1.21 ^{b, B}	7,66	0,123	< 0.001	0,04
	Placebo Group	31.9 ± 3.6 ^{a, A}	11,39	27 ± 3.89 ^{a, B}	15,46			
BUN	NAC Group	103.5 ± 4.79 ^{a, A}	30,3	51.45 ± 3.93 ^{b, B}	24,83	0,018	< 0.001	0,001
	Placebo Group	104.6 ± 8.33 ^{a, A}	26,33	91.4 ± 12.17 ^{a, A}	38,49			
CREA*	NAC Group	8.04 ± 0.35 ^{a, A}	2,19	4.01 ± 0.25 ^{b, B}	1,55	0,119	< 0.001	< 0.001
	Placebo Group	7.59 ± 0.58 ^{a, A}	1,82	6.44 ± 0.9 ^{a, B}	2,83			
PHOS	NAC Group	9.62 ± 0.6 ^A	3,77	6.28 ± 0.31 ^B	1,99	0,945	< 0.001	0,399
	Placebo Group	8.94 ± 0.98 ^A	3,11	6.85 ± 0.62 ^B	1,95			
TP	NAC Group	8.16 ± 0.15 ^A	0,94	7.74 ± 0.09 ^B	0,58	0,119	0,005	0,921
	Placebo Group	8.54 ± 0.25 ^A	0,79	8.08 ± 0.2 ^B	0,63			
ALB	NAC Group	3.11 ± 0.07	0,43	2.99 ± 0.05	0,35	0,864	0,218	0,51
	Placebo Group	2.87 ± 0.06	0,2	3.01 ± 0.11	0,34			
UPC	NAC Group	0.96 ± 0.07 ^{b, A}	0,41	0.52 ± 0.04 ^{b, B}	0,25	< 0.001	< 0.001	0,004
	Placebo Group	1.44 ± 0.18 ^{a, A}	0,57	1.33 ± 0.18 ^{a, A}	0,55			

RBC Red blood cell (10⁹/l), HCT Hematocrit (%), HGB Hemoglobine (g/dl), SDMA Symmetric Dimethylarginine (μg/dl), CREA Creatinine (mg/dl), BUN Blood urea nitrogen (mg/dl), PHOS phosphorus (mg/dl), TP Total protein (g/dl), ALB Albumine (g/dl), UPC Urine protein/creatinine ratio. *All cats in groups have ≥ 1.8 mg/dl creatinine levels

SEM Standard error of mean, SD Standard deviation

t0 Before treatment, t8 after treatment

^{a, b} Different letters in the same column for each variable indicate a statistically significant difference ($p < 0.05$)

^{A, B} Different letters in the same row for each variable indicate a statistically significant difference ($p < 0.05$)

helping to stabilize glucose levels in diabetic cats [13], helping to increase the total leukocyte count in parvovirus-infected dogs [33], and the ability of NAC to disrupt biofilm formation in dogs with otitis media as positive therapeutic effects [34]. Notwithstanding the plethora of studies conducted in the domain of veterinary medicine, the preponderance of research assessing the effects of NAC on kidney function has been confined to experimental animal models [35–37]. To the best of the author's knowledge, there have been no clinical studies conducted to evaluate the renoprotective effect of NAC in cats with naturally occurring chronic kidney disease. The objective of the present study was to evaluate the therapeutic effects of NAC in cats with acute-on-chronic kidney disease.

A significant corpus of human research has clearly demonstrated the relationship between chronic kidney dysfunction and oxidative stress and lipid peroxidation [38, 39]. In a study of rats that had undergone nephrectomy, a reduction in dietary antioxidant levels was found to accelerate the progression of kidney disease [40]. The precise mechanisms by which oxidative stress contributes to the development of chronic kidney dysfunction remain to be fully elucidated. However, an experimental study has demonstrated that levels of malondialdehyde (MDA) and lipid peroxidation products have been observed to increase in rats following 5/6 nephrectomy [1]. The administration of NAC has been demonstrated to reduce MDA levels and improve ischaemic kidney injury in haemodialysis patients [41]. It has been demonstrated that NAC administration engenders a renal protective effect by increasing glomerular filtration rate (GFR) and reducing proteinuria [42]. In addition, NAC application in rats has been shown to cause a progressive increase in GFR [42]. As demonstrated in the research, SDMA reflects early GFR declines before creatinine increases in cats with chronic kidney disease [43]. In this study, it was observed that the decrease in SDMA in cats with chronic kidney disease treated with NAC was statistically more significant when compared to the placebo group. This finding suggests that the addition of NAC to the intravenous fluid treatment resulted in a bigger improvement in GFR biomarkers than fluid therapy alone. Conversely, a study conducted on humans who naturally developed chronic kidney dysfunction found that NAC use reduced the likelihood of developing end-stage kidney disease by 18% [12]. The administration of NAC to patients suffering from chronic kidney dysfunction was conducted both prior to and following exposure to radiocontrast material. This resulted in a 19% reduction in contrast-induced nephropathy, as evidenced by a decline in serum creatinine levels [14]. Furthermore, NAC administration was found to decrease creatinine and BUN levels, in addition to proteinuria [44]. These data are consistent with

the findings of the present study. In this study, statistically significant decreases were observed in BUN, creatinine and UPC values in patients who received NAC compared to those who received a placebo. Despite the absence of a comprehensive understanding of the pathophysiological mechanisms through which NAC prevents the progression of chronic kidney dysfunction by reducing serum creatinine and BUN levels, it is hypothesised that the renoprotective property of NAC is related to the increase in glutathione synthesis capacity [45]. NAC has been demonstrated to offer a degree of protection to kidney tubules, in part due to its antioxidant properties, and has been observed to reduce kidney cell apoptosis and support cell repair [46]. In certain studies, NAC has been shown to specifically prevent mitochondrial dysfunction in cases of acute kidney injury, with other studies demonstrating its capacity to reduce mitochondrial damage caused by oxidative stress [47]. It can be extrapolated from the evidence presented that NAC administration may be beneficial in cats with acute kidney failure or acute exacerbations of chronic kidney disease [48]. Regulation of inflammatory and mitochondrial processes by NAC has also been demonstrated to be effective in preventing kidney fibrosis [13]. It is imperative to acknowledge the pivotal role of chronic inflammatory reaction in the progression of CKD. A study has demonstrated that NAC is capable of reducing inflammatory cell infiltration and pathological damage to kidney tissue by significantly decreasing IL6 [49]. The kidney-protective effect of NAC use in rats with renal tissue damage has also been proven through histopathological analysis [35]. The current literature reveals that antioxidant capacity is reduced in felines afflicted with chronic kidney disease when contrasted with those of healthy animals [32, 50]. A study found measuring oxidative damage markers like MDA more sensitive than other plasma markers for evaluating patients with chronic kidney dysfunction [3]. The determination of oxidative stress markers and antioxidant capacity in cats with chronic kidney disease was not among the objectives of the present study. Consistent with prior reports [51, 52], the provision of nutritional support is important in the management of cats with CKD and should be addressed in future prospective studies to ensure that cats receive adequate enteral nutritional whilst hospitalized. The absence of kidney histopathology and the inclusion of cats at different stages of chronic kidney disease in the study can be considered as limiting factors. Larger number of cats with CKD could be useful to further evaluation.

Conclusion

In conclusion, the findings of this study demonstrate that NAC was well tolerated and led to a significant reduction in creatinine, BUN, SDMA, and proteinuria within one

week compared to the placebo group. The incorporation of NAC into treatment regimens demonstrates potential as a treatment strategy in cats with acute-on-chronic kidney disease.

Acknowledgements

Not applicable.

Authors' contributions

Hadi Alihosseini: Conceptualization (lead); data curation (equal); investigation (equal); methodology (lead); writing – original draft (lead), validation (equal) Ekrem Çağatay Çolakoğlu: Conceptualization (lead); data curation (equal); investigation (equal); methodology (lead); writing – original draft (lead), validation (equal) Ali Evren Haydardedeoğlu: Data curation (equal); investigation (equal); methodology (lead); validation (equal). Doğan Özgen: Writing– review & editing, Methodology, Investigation, Formal analysis.

Funding

The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Data availability

The datasets generated and/or analysed during the current study are not publicly available but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Informed consent was obtained from all owners included in the study. The present study was carried out in accordance with the principles of the European Convention for the Protection of Vertebrate Animals Used for Experimental and Other Scientific Purposes and the requirements of European Union Directive 2010/63 on the protection of animals used for experimental and other scientific purposes. The authors confirm that the ethical policies of the journal, as noted on the journal's author guidelines page, have been adhered to and the appropriate ethical review committee approval has been received. The authors confirm that they have adhered to either US or European standards for the protection of animals used for scientific purposes.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹Therapist Veterinary Clinic, Istanbul, Turkey

²Department of Internal Medicine, Faculty of Veterinary Medicine, Small Animal Veterinary Teaching Hospital, Ankara University, Ankara, Turkey

³Department of Internal Medicine, Faculty of Veterinary Medicine, Small Animal Veterinary Teaching Hospital, Aksaray University, Aksaray, Turkey

⁴Department of Biostatistics, Faculty of Veterinary Medicine, Ankara University, Ankara, Turkey

Received: 14 August 2025 / Accepted: 22 January 2026

Published online: 03 February 2026

References

- Brown CA, Elliott J, Schmiedt CW, et al. Chronic kidney disease in aged cats: clinical features, morphology, and proposed pathogenesis. *Vet Pathol*. 2016;53:309–26.
- Himmelfarb J. Relevance of oxidative pathways in the pathophysiology of chronic kidney disease. *Cardiol Clin*. 2005;23:319–30.
- Locatelli F, Canaud B, Eckardt KU, et al. Oxidative stress in end-stage renal disease: an emerging threat to patient outcome. *Nephrol Dial Transpl*. 2003;18:1272–80.
- Small DM, Coombes JS, Bennett N, et al. Oxidative stress, anti-oxidant therapies and chronic kidney disease. *Nephrol Ther*. 2012;17:311–21.
- Duni A, Liakopoulos V, Roumeliotis S, et al. Oxidative stress in the pathogenesis and evolution of chronic kidney disease: untangling ariadne's thread. *Int J Mol Sci*. 2019;20:3711.
- Keegan RF, Webb CB. Oxidative stress and neutrophil function in cats with chronic renal failure. *JVIM*. 2010;24:514–9.
- Samuni Y, Goldstein S, Dean OM, et al. The chemistry and biological activities of N-acetylcysteine. *Biochim Biophys Acta*. 2013;1830:4117–29.
- Zafarullah M, Li WQ, Sylvester J, et al. Molecular mechanisms of N-acetylcysteine actions. *Cell Mol Life Sci*. 2003;610:6–a20.
- Aboubakr M, Elmahdy AH, Taima SH, et al. Protective effects of n acetylcysteine and vitamin e against acrylamide-induced neurotoxicity in rats. *Pak Vet J*. 2013;43:262–8.
- Andreucci M, Faga T, Serra R, et al. Update on the renal toxicity of iodinated contrast drugs used in clinical medicine. *Drug Health Patient Saf*. 2017;22:25–37.
- Gong X, Duan Y, Zheng J et al. Nephroprotective effects of N-acetylcysteine amide against contrast-induced nephropathy through upregulating Thioredoxin-1, inhibiting ASK1/P38MAPK pathway, and suppressing oxidative stress and apoptosis in rats. *Oxid Med Cell Longev*. 2016;1:8715185.
- Liao CY, Chung CH, Wu CC, et al. Protective effect of N-acetylcysteine on progression to end-stage renal disease: necessity for prospective clinical trial. *Eur J Intern Med*. 2017;44:67–73.
- Wang X, Bi X, Yang K. Et al. ROS/ P38MAPK-induced lamin B1 accumulation promotes chronic kidney disease-associated vascular smooth muscle cells senescence. *Biochem Biophys Res Commun*. 2020;531:187–94.
- Tepe M, van der Schwarzfeld GM. Prevention of radiographic-contrast-agent-induced reductions in renal function by acetylcysteine. *N Engl J Med*. 2020;343:180–84.
- Efrati S, Berman S, Siman-Tov Y, et al. N -acetylcysteine attenuates NSAID-induced rat renal failure by restoring intrarenal prostaglandin synthesis. *Nephrol Dial Transpl*. 2007;22:1873–81.
- Elliott J, Watson A. Chronic kidney disease: international renal interest society staging and management. In: Bonagura DT: current veterinary therapy. 2nd ed. New York: Saunders-Elsevier; pp 857–63.
- Acierio MJ, Brown S, Coleman AE, et al. ACVIM consensus statement: guidelines for the identification, evaluation, and management of systemic hypertension in dogs and cats. *JAVNU*. 2020;12:30–49.
- Plumb DC. *Veterinary Drug Handbook*. 7th edition, Minnesota, PharmaVet Inc; 2011.
- Wisner T. editor. Hepatic toxins and the emergent patient. IVECC Symposium. Veterinary Information Network; 2006 Available from <http://goo.gl/SbjSx>.
- Dibartola SP, Fluid. Electrolyte and acid base disorders in small animal practice. New York: Elsevier; 2006.
- Muir WW, DiBartola SP. Fluid therapy. Kirk RW: current veterinary therapy. Philadelphia: WB Saunders; 1983. p. 33.
- Dumville JC, Hahn S, Miles JNV, et al. The use of unequal randomisation ratios in clinical trials: A review. *Contemp Clin Trials*. 2006;27:1–12.
- Hey SP, Kimmelman J. Are outcome-adaptive allocation trials ethical? *Clin Trials*. 2014;11:565–71.
- Faul F, Erdfelder E, Lang AG, et al. GPower 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behav Res Methods*. 2007;39:175–91.
- Pedre B, Barayeu U, Ezerin AD, et al. The mechanism of action of n-acetylcysteine (nac): the emerging role of h2s and sulfane sulfur species. *Pharmacol Ther*. 2021;228:107–916.
- Tieu S, Charchoglyan A, Paulsen L, et al. N-Acetylcysteine and its Immunomodulatory properties in humans and domesticated animals. *Antioxidants*. 2023;12:1867.
- Buur JL, Diniz PPVP, Roderick KV, et al. Pharmacokinetics of n-acetylcysteine after oral and intravenous administration to healthy cats. *Am J Vet Res*. 2013;74:290–3.
- Petkova T, Milanova A. Absorption of n-acetylcysteine in healthy and Mycoplasma gallisepticum-infected chickens. *Vet Sci*. 2021;8:244.
- Meadows I, Editor. The 10 Most Common Toxicoses in Dogs. *Vet Med-Bonn Springs*; 2006. https://www.aspcapro.org/sites/default/files/zm-vetm0306_14-2-148.pdf.
- Ploneczka-Janeczko K, Lis P, Bierowiec K, et al. Identification of bap and IcaA genes involved in biofilm formation in coagulase negative Staphylococci isolated from feline conjunctiva. *Vet Res Commun*. 2014;38:337–46.

31. Hou Y, Wang L, Yi D, et al. N-acetylcysteine and intestinal health: a focus on its mechanism of action. *Front Biosci.* 2015;20:872–91.
32. Constantin NT. The effect of intra-uterine treatment with diluted n-Acetylcysteine on bovine endometritis. *C Vet Med.* 2018;64:31–3.
33. Gaykwad C, Garkhal J, Chethan GE, Et. Amelioration of oxidative stress using N-acetylcysteine in canine parvoviral enteritis. *J Vet Pharmacol Ther.* 2018;41:68–75.
34. May ER, Conklin KA, Bemis DA. Antibacterial effect of N-acetylcysteine on common canine otitis externa isolates. *Vet Dermatol.* 2016;27:18–47.
35. Dong G, Li Q, Yu C, et al. N-Acetylcysteine protects against diazinon-induced histopathological damage and apoptosis in renal tissue of rats. *Toxicol Res.* 2024;40:285–95.
36. Dong W, Zhang K, Gong Z, et al. N-acetylcysteine delayed cadmium-induced chronic kidney injury by activating the Sirtuin 1-P53 signaling pathway. *Chem Biol Interact.* 2023;369:110–299.
37. Fathi SM, Desouky A, Hussein SA. Et al. The protective effects of Lemongrass and N-acetylcysteine against Adenine-Induced chronic kidney disease in an animal model. *BVMJ.* 2024;47:22–7.
38. Annuk M, Zilmer M, Lind L, et al. Oxidative stress and endothelial function in chronic renal failure. *J Am Soc Nephrol.* 2001;12:2747–52.
39. Vaziri ND, Dicus M, Ho ND. Oxidative stress and dysregulation of superoxide dismutase and NADPH oxidase in renal insufficiency. *Kidney Int.* 2003;63:179–85.
40. Nath KA, Salahudeen AK. Induction of renal growth and injury in the intact rat kidney by dietary deficiency of antioxidants. *J Clin Invest.* 1990;86:1179–92.
41. Trimarchi T, Mongitore MR, Baglioni P, et al. N-acetylcysteine reduces malondialdehyde levels in chronic Hemodialysis patients— A pilot study. *Clin Nephrol.* 2023;99:441–6.
42. Heloisa M, Shimizu M, Coimbra TM, et al. N-acetylcysteine attenuates the progression of chronic renal failure. *Kidney Int.* 2005;68:2208–17.
43. Hall JA, Yerramilli MVS, Obare E, et al. Comparison of serum concentrations of symmetric dimethylarginine and creatinine as kidney function biomarkers in cats with chronic kidney disease. *JVIM.* 2014;28:1676–83.
44. Efrati S, Averbukh M, Berman S, et al. N-acetylcysteine ameliorates lithium-induced renal failure in rats. *Nephrol Dial Transpl.* 2005;20:65–70.
45. Aparicio-Trejo OE, Reyes-Fermin LM, Briones-Herrera A, et al. Protective effects of N-acetyl-cysteine in mitochondria bioenergetics, oxidative stress, dynamics and S-Glutathionylation alterations in acute kidney damage induced by folic acid. *Free Radic Biol Med.* 2019;130:379–96.
46. Xing Y, Wei RB, Tang L, et al. Protective effect of Salidroside on contrast-induced nephropathy in comparison with N-acetylcysteine and its underlying mechanism. *Chin J Integr Med.* 2015;21:266–73.
47. Kandel R, Singh KP. Higher concentrations of folic acid cause oxidative stress, acute cytotoxicity, and long-term fibrogenic changes in kidney epithelial cells. *Chem Res Toxicol.* 2022;35:2168–79.
48. Diehl SH, Seshadri R. Use of continuous renal replacement therapy for treatment of dogs and cats with acute or acute-on-chronic renal failure: 33 cases (2002–2006). *JVECC.* 2008;18:370–82.
49. He G, Li Q, Li W. N-acetylcysteine for preventing of acute kidney injury in chronic kidney disease patients undergoing cardiac surgery: a meta-analysis. *Heart Surg Forum.* 2018;21:513–21.
50. Chen H, Segev G. Evaluation of oxidative stress in dogs and cats with chronic kidney disease. *JVIM.* 2024;38:3105–10.
51. Sparkes AH, Caney S, Chalhoub S, et al. ISFM consensus guidelines on the diagnosis and management of feline chronic kidney disease. *JFMS.* 2016;18:219–39.
52. Parker VJ. Nutritional management for dogs and cats with chronic kidney disease. *Vet Clin North Am Small Anim Pract.* 2021;51:685–710.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.